Mohit Dubey

AI Full Stack Engineer

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SUMMARY

Senior Full Stack Developer with 5+ years of experience designing and building high-performance **React** applications and **Python/Node.js** backend services, paired with deep expertise in **AI agent** and **LLM application** development end-to-end — including agentic orchestration logic (**LangChain**, **LangGraph**, **ReAct**-style reasoning/action loops). Skilled in **REST/GraphQL API integration**, agent tooling, and **observability/tracing** for production agent systems. Comfortable owning architecture end-to-end in **Agile/Scrum** teams, with hands-on experience on the **Databricks** platform and **cloud-native DevOps** (AWS, Kubernetes, Docker, CI/CD, Infrastructure-as-Code).

WORK EXPERIENCE

Software Engineer —— Publicis Sapient June 2025 - Present

- Led the architecture and development of an enterprise-grade **GraphRAG**-based private coding assistant for a B2B SaaS platform using **LangGraph**, **DeepAgents**, and agentic AI, improving code retrieval accuracy by 38% and reducing developer resolution time by 60%.
- Designed and implemented an **MCP server** with dynamic tool discovery, semantic search, and context orchestration, reducing LLM context overhead by 70% and lowering token consumption and inference costs across production workloads.
- Built a **multi-tenant, multi-agent AI platform** with real-time guardrails, memory management, and policy enforcement, enabling secure deployment across **multiple tenants** while maintaining sub-second response latency.
- Architected scalable agent workflows leveraging **LangGraph**, vector databases, reranking, and hybrid retrieval techniques, increasing answer relevance by 42% and supporting thousands of concurrent enterprise requests.
- Continuously experimented with a broad portfolio of **Generative AI** model families — transformers, LLMs, and multimodal models — evaluating emerging AI tooling and agent frameworks for production readiness, security, and compliance before adoption.
- Co-developed features and integrations side-by-side with **GenAI coding agents** (agentic orchestration, MCP tool-calling), cutting time-to-market on new capabilities and shipping on a rapid, near-continuous release cadence with end-to-end ownership.
- Built high-performance, scalable **REST APIs** integrating both documented and undocumented internal and third-party service interfaces, adhering to security best practices and secured via **OAuth2**.
- **Partnered with executive leadership** to identify high-impact AI opportunities, quantify ROI, and drive new revenue streams from tech innovation.
- Collaborate closely with Product, Engineering, and Data Governance groups to enforce strict frameworks for **ethical AI, regulatory compliance, and responsible model deployment**.
- Established **DevOps** practices (CI/CD, containerized deployments, automated tracing and monitoring) for agent platform releases, enabling a rapid, near-continuous release cadence.
- Managed the full project delivery lifecycle from conception to production rollout, maintaining a consistent **95% project success rate**.
- Implemented **performance-optimization** techniques such as **code-splitting, lazy loading, and dynamic imports** to reduce **JavaScript bundle size** and improve time to interactive (TTI).
- Engineered scalable **Frontend Architecture** using **React.js**, **Next.js**, and micro-frontends, optimizing platform performance by **60%** and securing workflows using **OAuth2** authentication protocols.

Software Engineer —— Deqode Aug 2023 - May 2025

- Built real-time streaming pipelines with **Apache Flink** and Kafka, and large-scale batch processing on **Hadoop** (HDFS), feeding document ingestion and feature pipelines for the RAG platform.
- Containerized services with **Docker** and established **DevOps CI/CD pipelines** with automated testing and staged rollouts, eliminating manual deployment errors and enabling zero-downtime releases.
- Architected enterprise-grade **RAG pipeline** using **LangChain**, Pinecone/FAISS, and AWS, reducing LLM hallucinations by 32% and improving retrieval latency by 45% across 1M+ documents.
- Fine-tuned 7B/13B **Qwen3-VL** LLMs using **SFT** and **QLoRA** on **AWS SageMaker** GPU clusters, improving domain-specific accuracy by 21% while cutting inference cost by 60%.
- Deployed production **LLM serving infrastructure** on **GCP Vertex AI** with **vLLM** and TensorRT-LLM, serving 13B parameter models at sub-200ms latency with 40% cost reduction for 500K+ requests/day.
- Orchestrated cloud-native ML systems on **Kubernetes** with **MLflow**, Terraform, and Vertex AI Pipelines using **DevOps**/MLOps best practices; led platform modernization for **15+ ML teams**, compressing model deployment cycle from 3 weeks to 2 days.
- Architected **distributed training** using DeepSpeed and FSDP for multi-GPU foundation model runs, reducing training time by 65% and scaling to 34B+ parameter models.
- Deployed fluid, multi-agent conversational systems leveraging LangChain Agents, Streamlit UIs, and AWS Bedrock API gateways; **mentored AI engineering teams** on RAG scaling laws and context-window optimization.
- Created robust **AI monitoring and real-time dashboarding infrastructures** using Grafana, Power BI, and Looker to continuously track operational metrics, system health, and model drift.

- Translated deep AI behavioral insights into predictive analytics models, enhancing customer experience metrics and optimizing down-stream system performance.
- Built Aadhaar-Voter ID linking UI in React using **Bootstrap** for responsive layouts, scaling to **handle high-traffic production loads** with resilient, reusable components.
- Migrated **Redux** to modular **RTK** slices + RTK Query, reducing API overfetching by 78%.
- Developed editable virtualized grid (1M+ rows) using **react-window**, with smooth scroll and memoization.
- Led federated design **system adopted by 50+ teams**, enforced standards.
- Integrated Web Vitals + **Sentry** to track LCP, FCP, TTI across 500k+ sessions/month and optimize core web metrics.
- Built **secure file uploader** with presigned S3, retry/resume, chunked upload, drag-and-drop UI, and Fetch API.
- Implemented **keyboard navigation**, focus management, ARIA roles, and screen reader support using semantic HTML.
- Built schema-to-React renderer supporting **dynamic layouts**, conditional fields, async data bindings, and versioned JSON configs; cut Ul rollout time.

Software Engineer —— Shiv Technolabs Feb 2023 - July 2023

- **Leveraged deep knowledge of browser internals** to optimize performance, systematically reducing reflow,repaint, and bundle sizes to achieve significant improvements in Core Web Vitals.
- **Implemented advanced state management** patterns across complex application workflows using MobX and modern ES6+ JavaScript features for predictable data flow.
- **Led frontend performance audits** analyzing JS engines compilation behaviors and execution contexts to eliminate memory leaks and runtime bottlenecks.
- Orchestrated the transition to **Next.js** methodologies, integrating **RESTful APIs** to achieve seamless data transfer across platforms, which decreased website bounce rate by **15%** and grew traffic.

Software Engineer —— Sourcebae May 2021 - Jan 2023

- Built full-stack applications with **React**, **TypeScript**, and **NestJS**, shipping REST API backends backed by **PostgreSQL** and **MongoDB** serving 100K+ users with sub-100ms p99 latency via Redis caching.
- Designed a reusable **TypeScript** component library with centralized state management, cutting feature delivery time by 40% across 5 concurrent product teams.
- Containerized services with **Docker** and established **CI/CD pipelines** with automated testing and staged rollouts, eliminating manual deployment errors and enabling zero-downtime releases.
- Built **offline-first UI** in React Native using SQLite for local data storage and background sync to update the server when reconnected.

PROJECTS

TradeBrains — Automated Real-Time Algorithmic Trading Platform

- Architected an end-to-end algorithmic trading platform predicting 20-minute stock movements by fine-tuning open-source generative models (**Llama 3.3 70B**) using 3-stage **RLHF** (**SFT**, **Reward Model**, **PPO**) and **GRPO** to align prediction outputs directly with target performance metrics.
- Implemented custom **GRPO/PPO reward functions** integrating multi-objective metrics (factual correctness, formatting compliance, and financial risk metrics) to eliminate hallucinations in high-stakes domain outputs.
- Engineered **QLoRA 4-bit** parameter-efficient fine-tuning (**PEFT**) pipelines for 8B-parameter sentiment models, reducing memory footprints while maintaining **88% structured score extraction accuracy**.
- Executed **Direct Preference Optimization (DPO)** and **KTO** fine-tuning on domain-specific instruction datasets, bypassing separate reward model latency while matching RLHF alignment quality.

EDUCATION

Rajiv Gandhi Proudyogiki Vishwavidyalaya Bhopal, India

Bachelor of Engineering (HONOURS) - Information Technology; GPA: 7.75 August 2016 - May 2020

ACHIEVEMENTS

- **AWS Certified Developer Associate**.
- **AWS Certified AI Practitioner**.
- **AWS Certified Cloud Practitioner**.
- **Hacker Rank Javascript Basic**.
- **Hacker Rank CSS certification**.
- **Sololearn React + Redux**

TECHNICAL SKILLS

- **LLMs & Agents**: LangChain, LangGraph, CrewAI, Google ADK, LlamaIndex, Hugging Face Transformers, OpenAI API, Ollama, vLLM, TensorRT-LLM, MCP Servers, Agent Orchestration, RAG Systems, Prompt Engineering
- **Training & Optimization**: LLM Fine-Tuning (SFT, QLoRA, LoRA, PEFT), RLHF, GRPO, DeepSpeed, FSDP, Distributed Training, CUDA, Model Quantization (INT8, NF4)
- **ML & Deep Learning**: PyTorch, TensorFlow, scikit-learn, XGBoost, Classical ML, Feature Engineering, Model Optimization

- **Data Engineering**: Apache Spark (PySpark, Spark SQL), Apache Flink, Hadoop, Databricks, Delta Lake, Kafka, Airflow, BigQuery, ETL/ELT, Feature Store (Feast)
- **MLOps & Infrastructure**: DevOps, MLflow, Kubeflow, Kubernetes, Docker, Terraform, CI/CD Pipelines, Model Monitoring, Drift Detection, A/B Testing
- **Cloud**: AWS (SageMaker, EKS, Lambda, S3), GCP (Vertex AI, GKE, Dataproc, BigQuery)
- **Backend Engineering**: Python, FastAPI, Go, TypeScript, REST APIs, gRPC, Microservices, WebSocket, Redis, n8n
- **Databases & Vector Stores**: PostgreSQL, MongoDB, Redis, Qdrant, Milvus, Weaviate, Chroma, SQL
- **Frameworks:** Reactjs, Redux, Nextjs, SCSS, MaterialUI, ExpressJs, Jest, RTL, Enzyme
- **Tools:** Azure, Docker, Webpack, Git/Github, Postman, Swagger, Figma, Sequelize ORM, Firebase
- **Coursework:** System Design, Database Management Systems, Data Structures and Algorithms, Object Oriented Programming
- **Other:** RESTful Web Services, Agile methodologies, Integration testing, micro frontends, Incident Response & Operational Excellence